10.6 Two Bodies Become One

WORKSHEET

The question

STEP III 2004, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q11

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Write down Newton's second law for each particle. In which direction does each force act?

STAGE 1 The separation

Ask yourself Which combination of the two equations involves only z?

STAGE 2 Energy for the separation

Ask yourself With $\ddot{z} = v\frac{\mathrm{d}v}{\mathrm{d}z}$, separate the variables and integrate. What are the initial values of z and $v = \dot{z}$?

STAGE 3 A surprisingly simple solution

Ask yourself The answer to show is $z^2 = 4t^2 - 2t + 1$. So look at $w = z^2$ instead of z. What is $\dot{w}^2$ in terms of w?

STAGE 4 **The centre of mass**

Ask yourself What equation does $2x_1 + x_2$ satisfy? Why is it the natural combination to look at?

What's really going on

Why z^2 is quadratic

For any force law of the form $\ddot{z} = k/z^3$, the same calculation gives $\frac{d^2}{dt^2}(z^2) = 2\dot{z}^2 + 2z\ddot{z} = 2\dot{z}^2 + \frac{2k}{z^2} =$ constant, since $\frac{1}{2}\dot{z}^2 + \frac{k}{2z^2}$ is the conserved energy. So z^2 is always a quadratic in t. Why?

Here is the hidden reason. Think of a particle moving freely, in a straight line at constant speed, in a *plane*. Its distance r from the origin satisfies $r^2 =$ (a quadratic in t), since $r^2 = (x_0 + at)^2 + (y_0 + bt)^2$. And in polar coordinates its radial equation of motion is $\ddot{r} - r\dot{\theta}^2 = 0$; with the angular momentum $h = r^2\dot{\theta}$ constant, that becomes

$$\ddot{r} = \frac{h^2}{r^3}.$$

The inverse-cube repulsion is exactly the *centrifugal* effect of the hidden angular motion. In our problem, $z^2 = 4t^2 - 2t + 1 = (2t - \frac{1}{2})^2 + \frac{3}{4}$ is the squared distance from the origin of a point moving at speed 2 along a straight line that passes the origin at distance $\frac{\sqrt{3}}{2}$. The one-dimensional repelling particles are the shadow of free motion in two dimensions.

The two-body problem

The method of this question works for any force between two bodies that depends only on their separation and acts along the line between them, in three dimensions as well as one. The centre of mass moves uniformly, and the separation vector $\mathbf{r}$ obeys $\mu\ddot{\mathbf{r}} = \mathbf{F}(\mathbf{r})$, with $\mu = \frac{m_1 m_2}{m_1+m_2}$. For gravity this is the *Kepler problem*, solved by Newton in the *Principia*: the separation traces an ellipse, parabola or hyperbola. The Earth and the Sun both orbit their common centre of mass, which lies inside the Sun; the Earth and the Moon orbit a point about 4700 km from the Earth's centre, inside the Earth but well away from its centre, which is why the Earth wobbles monthly. Astronomers find planets round other stars by detecting exactly this wobble in the star.

Integrable systems

In 1969 Francesco Calogero studied many particles on a line repelling each other with inverse-square potentials (that is, inverse-cube forces), and found that, remarkably, the system can be solved exactly for any number of particles. Jürgen Moser explained the reason in 1975, using a matrix whose eigenvalues stay constant as it evolves: the *Lax pairs* of Section 7.7. The *Calogero–Moser system* is now a central example of an *integrable system*, a system with enough conserved quantities to be solved completely, and it connects mechanics with random matrices, quantum field theory and the theory of solitons.

Try this next

1. **Two masses on a string.** Try STEP III 2023, Question 9 (in OCR's download of the 2023 papers, www.ocr.org.uk/Images/707102-2023-past-papers.zip). It follows two particles joined by a string, using the centre of mass to separate the motion. Look for when one particle moves backwards.

2. **Attraction.** Repeat the question with an attractive force $\frac{2}{z^3}$. Show that z^2 is still a quadratic in t, and find when the particles collide.

3. **The hidden plane.** For the motion in the question, find the straight line in the plane whose distance from the origin is $z(t)$, and the angle $\theta(t)$ swept out. Check that $z^2\dot{\theta}$ is constant and that $(z^2\dot{\theta})^2 = 3$.

4. **Kepler's third law.** Two stars of masses m_1 and m_2 move in circles about their centre of mass, a distance a apart, under gravity. Show that the period T satisfies $T^2 = \frac{4\pi^2 a^3}{G(m_1+m_2)}$.

Further reading

- **John Taylor, *Classical Mechanics*** (University Science Books), Chapter 8. The two-body problem, reduced mass and orbits.

- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), Chapter 10, on central forces.
- **Francesco Calogero, *Classical Many-Body Problems Amenable to Exact Treatments*** (Springer). For the ambitious: the world of exactly solvable systems.

The full guided solution is in Section 10.6:
https://beyondstep.keithmcnulty.org/ch10/two-body.html